股票交易系統技術報告

一、 使用策略

在多方嘗試後，最後我選擇了 RSI 策略(Relative Strength Index)，此策略又被稱為「相對強弱指標」，這個指標透過計算股票在一段時間內的收盤價，評估價格變動的速度。

RSI 公式：

$$\frac{A}{A+B} \times 100$$

A =一定期間上升幅度的合計
B =一定期間下跌幅度的合計

RSI 值越大，表示在過去一段期間，上漲機率較大；相反的，RSI 值越小，表示過去一段期間，下跌機率較大。RSI 指標的原發明者 Wilder 用 RSI 值的 70、30 做區分，當 RSI 值大於等於 70 時，表示價格被超買或高估；當 RSI 數值小於等於 30 時，表示價格被超賣或低估。指數越高代表市場越活絡，指數越低代表市場越低迷。

結論：當 RSI 值大於 70 則賣出；小於 30 則買進

二、 程式說明及策略回測

程式碼（僅擷取策略部分）：

```
class RSI_Strategy(Strategy):
    def init(self):
        self.rsi_period = 56 #設定 56 天計算一次 RSI 值
        self.gains = []
        self.losses = []
        self.avg_gain = 0
        self.avg_loss = 0

    def calculate_rsi(self):# 計算初始收益和損失
        for i in range(1, self.rsi_period + 1):
```

```
            price_diff = self.data.Close[i] - self.data.Close[i -
1]
            if price_diff >= 0:
                self.gains.append(price_diff)
                self.losses.append(0)
            else:
                self.losses.append(abs(price_diff))
                self.gains.append(0)
        # 計算平均收益和損失

        self.avg_gain = sum(self.gains) / self.rsi_period
        self.avg_loss = sum(self.losses) / self.rsi_period
    # 計算 RSI

        if self.avg_loss != 0:
            rs = self.avg_gain / self.avg_loss
            self.rsi = 100 - (100 / (1 + rs))
        else:
            self.rsi = 100
    # 更新收益和損失
    def next(self):
        price_diff = self.data.Close[-1] - self.data.Close[-2]
        if price_diff >= 0:
            self.gains.append(price_diff)
            self.losses.append(0)
        else:
            self.losses.append(abs(price_diff))
            self.gains.append(0)
    # 更新平均收益和損失

        self.avg_gain = (self.avg_gain * (self.rsi_period - 1) +
max(price_diff, 0)) / self.rsi_period
        self.avg_loss = (self.avg_loss * (self.rsi_period - 1) +
abs(min(price_diff, 0))) / self.rsi_period
        # 重新計算 RSI

        if self.avg_loss != 0:
            rs = self.avg_gain / self.avg_loss
            self.rsi = 100 - (100 / (1 + rs))
        else:
```

```
            self.rsi = 100
        # 基於 RSI 生成交易信號
        if self.rsi > 70:  #超賣條件
            self.sell()
        elif self.rsi < 30:  #超買條件
            self.buy()
```

回測結果：

最初我的 rsi_period 是設定 14，但回測結果 return 值會是負的

```
Start                     2017-01-02 02:00:00
End                       2022-12-30 16:58:00
Duration                   2188 days 14:58:00
Exposure Time [%]                    99.99991
Equity Final [$]                 29324.365797
Equity Peak [$]             100041923.010626
Return [%]                          -99.970676
Buy & Hold Return [%]                1.772032
Return (Ann.) [%]                  -66.449195
Volatility (Ann.) [%]                2.579555
Sharpe Ratio                              0.0
Sortino Ratio                             0.0
Calmar Ratio                              0.0
Max. Drawdown [%]                   -99.97075
Avg. Drawdown [%]                  -25.014874
Max. Drawdown Duration     2188 days 13:53:00
Avg. Drawdown Duration      547 days 03:44:00
```

所以改成 28，結果還是負的，但是 return 值有變大

```
Start                     2017-01-02 02:00:00
End                       2022-12-30 16:58:00
Duration                   2188 days 14:58:00
Exposure Time [%]                    99.99991
Equity Final [$]              19962761.493619
Equity Peak [$]             100147564.650661
Return [%]                          -80.037239
Buy & Hold Return [%]                1.772032
Return (Ann.) [%]                  -19.405275
Volatility (Ann.) [%]                5.358637
Sharpe Ratio                              0.0
Sortino Ratio                             0.0
Calmar Ratio                              0.0
Max. Drawdown [%]                  -81.643949
Avg. Drawdown [%]                   -9.085419
Max. Drawdown Duration     2188 days 13:25:00
Avg. Drawdown Duration      243 days 04:19:00
```

所以我再把時間加長，直接加到 56 天，return 值就變正的了

Start	2017-01-02 02:00:00
End	2022-12-30 16:58:00
Duration	2188 days 14:58:00
Exposure Time [%]	99.99991
Equity Final [$]	120166754.833327
Equity Peak [$]	122519945.527739
Return [%]	20.166755
Buy & Hold Return [%]	1.772032
Return (Ann.) [%]	2.48415
Volatility (Ann.) [%]	6.968496
Sharpe Ratio	0.356483
Sortino Ratio	0.531517
Calmar Ratio	0.174017
Max. Drawdown [%]	-14.275292
Avg. Drawdown [%]	-0.170282
Max. Drawdown Duration	1658 days 23:35:00
Avg. Drawdown Duration	3 days 02:11:00

之所以會有這個結果，可能是因為在考慮更長時間時，價格波動顯得較小，對於短時間內的價格波動也會較不敏感。